Control Plan No.: CP-001 | Rev: A | Date: 2026-11-05

Part / Assy: UVM-100 UV LED Optical Module | Phase: Pre-launch

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Linked documents: PFMEA-001, WI-001, UVM-001 Rev B, UVM-004 Rev B, FIX-001, GAG-100

Class: CC = critical, SC = significant (same as PFMEA-001)

Control Plan: UVM-100 Module Assembly

Op	Process step	Device / tool	Characteristic	Type	Class	Specification	Measurement method	Sample	Freq.	Control method	Reaction plan	PFMEA ref
10	Inspect housing	GAG-100	Seat depth (floor to ledge)	Product	CC	5.00 ±0.10 mm	Flush-pin gauge	100%	Every part	Go/no-go, result logged	Reject and quarantine housing; notify Quality	1
10	Inspect housing	GAG-100 + master part	Gauge reading on master	Process	CC	Pin top within step on master	Check gauge on master setting part	1	Start of shift	Gauge verification log	Stop; clean or replace pin; re-check all housings since last good check	2
20	Load housing	FIX-001 nest	Housing seated in nest	Process	—	Fully seated, protrudes 5 mm	Visual	100%	Every part	Per WI-001	Remove, clean pocket, reload	3
30	Install LED board	ESD mat, wrist strap	ESD protection active	Process	SC	Strap and mat pass resistance test	ESD tester	1	Start of shift	ESD check log	Stop; replace strap/mat; quarantine boards handled since last pass	5
30	Install LED board	FIX-001, key tab	Board orientation	Product	CC	Notch engaged on key tab	Poka-yoke (won’t seat otherwise)	100%	Every part	Error-proofed by design	Remove; inspect board for damage; reinstall	4
30	Install LED board	Air blow-off, magnifier	Board flat on floor	Product	SC	No gap or tilt	Visual under magnifier	100%	Every part	Per WI-001	Remove; clean floor; reinstall	6
40	Place lens	Inspection light	Lens orientation	Product	CC	Flat side down (orientation mark visible)	Visual	100%	Every part	Per WI-001	Remove; reorient; re-inspect	7
40	Place lens	Air blow-off	Lens seated	Product	SC	Rim flush and level on ledge	Visual	100%	Every part	Per WI-001	Remove; clean ledge; re-place	8
40	Place lens	Inspection light, finger cots, vacuum pen	Lens surface	Product	SC	No scratches, fingerprints or particles	Visual under inspection light	100%	Every part	WI-001 acceptance criteria	Clean if contaminated; scrap if scratched	9
50	Press retainer	Press tool FIX-002 (planned)	Retainer seated	Product	SC	Flush and level on lens rim, no gap	Visual	100%	Every part	WI-001 acceptance criteria	Re-press; if not seated, scrap retainer	10
50	Press retainer	Inspection light	Lens integrity	Product	CC	No cracks or chips	Visual under inspection light	100%	Every part	WI-001 acceptance criteria	Scrap module; notify Mfg Eng; check press force	11
60	Final inspection	Power supply, light check	Light output	Product	CC	LED lights, output present (planned)	Power-on check	100%	Every part	Pass/fail logged	Reject; quarantine; failure analysis	5, 12
60	Final inspection	Boundary samples	Overall workmanship	Product	CC	Matches boundary samples	Visual	100%	Every part	WI-001 acceptance criteria	Reject; route to rework or scrap	12

Items marked (planned) come from PFMEA-001 actions and must be in place before production release.